MEMS 1041

Final Report for Semester Team Project

Design and Construction of a Sensor Measuring Breakage Stress

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Abstract

A method to measure the strength of a replacement iPhone screen was determined. This study was conducted over the course of multiple lab periods. Because of this, the procedure was divided into different sections including designing and mounting the strain gauge, completing the circuitry, calibrating the sensors, and finally conducting the tensile strength test. Using strain gauge theory, Wheatstone bridge theory, signal conditioning, calibration techniques, and hypothesis testing, the team validated the design and testing results. Ultimately, it was found that the breakage stress of the replacement screen was 266.55 MPa which is below the expected value of 350 MPa ± 40 MPa by approximately 24%.

Objective

The overall objective of this project was to design a method of measuring the ultimate stress of an alternative replacement for an iPhone glass screen. By constructing this sensor and performing a ball drop test, the team collected data that can be compared to the expected strength provided in the manufacturer's specifications, which claim that these screens have the same strength as the original iPhone screens.

Theory

1. Stain Gage Theory

A strain gauge measures the deformation (strain) of a surface by detecting changes in its electrical resistance. It contains a thin metal wire in a serpentine pattern, whose resistance R will change when strained. By measuring ΔR we can infer the strain. The fundamental relationship is:

$$\frac{\Delta R}{R} = GF \cdot \varepsilon \qquad \text{[I]}$$

where

- ΔR = change in resistance [Ω]
- R = nominal resistance of the gage [Ω]
- GF = gage factor, which is constant over the operating range of the gage.
- ε = strain on the surface[m/m]

When the surface elongates, the gauge wire's length increases and cross-sectional area decreases, increasing its resistance proportionally. By relating strain to force through beam bending theory, the strain gauge output can quantify the applied load on the glass.

1.1. Beam Theory for Strain Gage Placement

The glass screen is modeled as a simply supported beam subjected to a central point load (the impact of the dropped mass).

For such a beam, the maximum bending moment M occurs at midspan and is given by:

$$M = \frac{P}{2} \cdot \frac{L}{2} \qquad \text{[II]}$$

where

- M = bending moment [Nm]
- P = point force by the dropped mass [N]
- L = distance between the supports [m]

The maximum bending stress is:

$$\sigma = \frac{M\frac{h}{2}}{I}$$

[III]

where

- σ = bending stress [Pa]
- h = glass thickness [m]
- $I = \frac{bh^3}{12}$ = moment of inertia for a rectangular cross-section beam $[m^4]$
- b = beam width [m]

The strain at the surface is related to stress via Hooke's Law:

$$\varepsilon = \frac{\sigma}{E} = \frac{3PL}{Ebh^2} \quad [IV]$$

where

- E = Young's modulus of the glass [Pa]

This equation allows prediction of strain at any location given an applied load, guiding gauge placement for maximum measurable strain without exceeding the gauge's limit(~4000με).

2. Wheatstone Bridge Theory

Because a single strain gauge produces a very small change in resistance, a Wheatstone bridge is used to convert resistance changes into measurable voltage differences.
The bridge output voltage for small strains is approximately:

$$E_o = E_i(\frac{\Delta R}{4R}) = E_i\frac{(GF)\varepsilon}{4} \quad [V]$$

where

- E_o = output voltage [V]
- E_i = input voltage [V]

Using a half-bridge with one active and one dummy gauge doubles the sensitivity and compensates for temperature effects.

3. Signal Conditioning

3.1. Frequency Response and Filter Classification

Fundamental behavior of a circuit is defined by the transfer function H(f), which describes the ratio of output to input voltage as a function of frequency:

$$H_f = \frac{V_{out}}{V_{In}} \quad [VI]$$

Filters are classified by which frequencies they pass:

- Low-pass filters allow frequencies below a cutoff frequency f_c to pass
- High-pass filters allow frequencies above a cutoff frequency f_c to pass

The frequency response is typically expressed in decibels:

$$Gain\ (dB)\ =\ 20\ *\ log\ (\frac{V_{out}}{V_{In}}) \quad [VII]$$

Where

- Gain = voltage gain decibels [dB]
- V_{out} = output voltage [V]
- V_{in} = input voltage [V]

3.2. Cutoff Frequency

The cutoff frequency f_c is where the gain drops by 3 dB from the passband level. For RC filters:

$$f_c = \frac{1}{2\pi RC} \quad [VIII]$$

Where

- f_c = cutoff frequency [Hz]
- R = resistance [Ω]
- C = capacitance [F]

3.3. Roll-Off Slope

Beyond the cutoff frequency, first-order filters attenuate at:

Roll-off slope = -20 dB/dec

3.4. Active Filter Gain

The active low-pass filter uses an op-amp in inverting configuration. The passband gain is:

$$G = \frac{R_2}{R_1} \quad [VIIII]$$

$$Gain\ (dB)\ =\ \frac{R_2}{R_1} \quad [X]$$

Where

- R_2 = feedback resistor [Ω]
- R_1 = input resistor [Ω]

3.5. Amplifier Theory

The differential amplifier is a combination of operational amplifiers and four resistors. This is designed to amplify the voltage difference between the two voltage inputs from the Wheatstone bridge. This output voltage can be calculated using the equation below:

$$V_1 = G_{amp}(V_A - V_B) \quad [XI]$$

$$G_{amp} = \frac{R_6}{R_5} \quad [XII]$$

It is important to use two resistors that have similar values for both R_5 and R_6 as this will prevent imbalance in the amplifier circuit.

3.6. Filter Design Theory

The active, low-pass filter is created using an operational amplifier, two resistors, and a capacitor. A filter is generally used to remove any unnecessary frequencies from a signal. An active filter is one that can provide amplification of a signal, thus requiring external power input for the op-amp. Determining a cutoff frequency for the filter is an important aspect of preventing aliasing. Usually, the cutoff frequency aims for over 20 dB signal reduction at the Nyquist frequency. The cutoff frequency of an active, low-pass filter is found using the equation:

$$f_c = \frac{1}{2*\pi*R*C} \qquad \text{[XIII]}$$

$$G_{filter} = \frac{R_8}{R_7} \qquad \text{[XIIII]}$$

4. Calibration Theory

Calibration is the process of determining the mathematical relationship between the applied force on the glass beam and the voltage output from the sensing system. In this project, the sensor system consists of:

a. The strain-gage instrumented glass beam
b. The Wheatstone bridge and amplifier circuit
c. The data-acquisition system used to measure the output voltage

During calibration, known forces are applied to the center of the beam by adding standard masses. For each mass m and gravitational acceleration g, the applied load F is:

$$F = mg \qquad \text{[XV]}$$

By holding each mass on the beam for a fixed duration, the system produces a near-constant output voltage. Averaging their constant regions gives one voltage value per applied force. Plotting V(output) vs F(input) and applying a linear fit yields the static sensitivity:

$$S = \frac{V}{F} \qquad \text{[XVI]}$$

This slope characterizes how much output voltage changes per unit of applied load and is used later to convert voltage recordings into actual force measurements.

5. Expected Sensitivity Theory

The applied force F at the beam center produces a predictable strain at the gage location. That strain produces a small bridge output voltage, which is then multiplied by the amplifier. Therefore, the theoretical sensitivity has the form:

$$\left(\frac{V}{F}\right)_{theoretical} = \left(\frac{V_{bridge}}{\varepsilon}\right)\left(\frac{\varepsilon}{F}\right)\left(G_{amp}\right) \quad \text{[XVII]}$$

Only the final slope is compared, not the intercept. Because the intercept depends strongly on zero-offset and bridge balancing, which vary from system to system.

6. Data Acquisition Theory

The DAQ is used to record voltage over time at a fixed sampling rate. Since the calibrated signals are relatively small, the input range must be chosen so that the maximum expected voltage fits within the DAQ's limits(±10V). The MATLAB script must:

a. Acquire voltage for a fixed time interval
b. Display a real-time plot of voltage vs. sample number/time
c. Save the data to a .mat workspace file

These saved datasets provide the voltage plateaus needed to calculate average voltage values for each applied mass.

7. Hypothesis Testing (t-test)

After testing and collecting multiple samples, the measured ultimate stresses are compared to the expected strength of standard glass. In order to account for any errors and uncertainty in the data collected from these experiments, hypothesis testing will be used. It is a statistical framework used to analyze sample data and compare it to assumptions. First, two hypotheses are made. The null hypothesis (H_o) assumes that the two data sets are indistinguishable. The alternative hypothesis (H_a) assumes that there is some statistically significant difference between the two data sets. Next, a significance level (α), which is the probability of being incorrect, is chosen. This value is typically 0.05 or 5%. Additionally, a critical value(t_{crit}) is determined to define the boundaries of the test using the t-distribution tables and the degrees of freedom (υ), which is calculated with the equation below:

$$DoF\ (\nu) = \frac{\left[\left(\frac{s_1^2}{n_1}\right)+\left(\frac{s_2^2}{n_2}\right)\right]^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1}+\frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}} \quad \text{[XVIII]}$$

where

- s_1 = sample standard deviation of data 1
- s_2 = sample standard deviation of data 2
- n_1 = sample size of data 1
- n_2 = sample size of data 2
    - If $n_2 \rightarrow \infty$, $v = n_1 - 1$.

A test statistic (t_{stat}) is calculated and compared to the critical value using the equation below:

$$t_{stat} = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{(\frac{s_1}{n_1})^2 + (\frac{s_2}{n_2})^2}} \quad \text{[XIX]}$$

where

- $\overline{x}_1$ = mean of sample data 1
- $\overline{x}_2$ = mean of sample data 2
- s_1 = standard deviation of data 1
- s_2 = standard deviation of data 2
- n_1 = sample size of data 1
- n_2 = sample size of data 2

A one-tailed test checks for an effect in one direction. Therefore, if there is a positive t_{stat} value ($\overline{x}_1 > \overline{x}_2$), a right-tailed test must be used. If there is a negative t_{stat} value ($\overline{x}_1 < \overline{x}_2$), a left-tailed test must be used. This comparison will determine if the null hypothesis can be accepted or rejected. If this value falls in the critical region, the null hypothesis is rejected. Otherwise, the null hypothesis is accepted. A two-tailed test checks if there are effects in any direction. This test does not specify which direction has an effect, so it can be used to determine if there is simply a difference between the test samples. This test is more conservative than the one-tailed test. The null hypothesis for a two-tailed test is simply that $\mu_1 \neq \mu_2$. However, the null hypothesis in a one-tailed test is either $\mu_1 > \mu_2$ or $\mu_1 < \mu_2$.

Procedure

Gauge Placement Design Strategy
This lab involves designing the optimal placement of strain gauges on a glass specimen that will be subjected to impact testing.

Gage Placement Design

Initial gage placement design: A half-bridge configuration was selected for this experiment, with two strain gages mounted on the top surface of the glass. The gauges were positioned horizontally with their midpoints approximately 30mm from the center impact zone, maintaining equidistant separation from the expected drop point.00 mm width. The orientation was to have the two gauges horizontally aligned.

Strain Calculation Model

An Excel spreadsheet was developed to calculate the expected strain in the gauges for a given input force. The glass specimen dimensions were measured: thickness (h) = 0.88mm, width (b) = 64mm, and the distance between supports (L_bend) = 100mm. Young's modulus for Corning Gorilla Glass (E = 77 GPa) was obtained from the manufacturer's specification sheet. The strain at each gauge location was calculated using Equation 6 from the theory section. Initial validation of the spreadsheet was performed using an assumed input force of 10N at a distance of 30mm from the support, which yielded an expected strain of approximately 240 microstrain, consistent with the expected range of 230-250 microstrain.

Breaking Force Determination

The force required to fracture the glass was calculated by rearranging Equation 7 to solve for the input force P. The ultimate stress of the glass was taken as 350 MPa ± 40 MPa as specified in the design requirements. Since maximum stress occurs at the center of the beam where the force is applied, the distance x was set to half the span length (50mm). This calculation was added to the Excel spreadsheet to determine the breaking force.

Strain Verification at Breaking Point

The breaking force calculated in the previous step was substituted into Equation 6 to determine the expected strain in the gages at the moment of glass fracture. This strain value was compared against the operational limits of the strain gages (500-4000 microstrain) to verify that the gage placement was appropriate. If the calculated strain exceeded 4000 microstrain or fell below 500 microstrain, the gauge placement would need to be adjusted and the calculations repeated.

Drop Height Calculation

The height from which the 0.21 kg mass must be dropped to break the glass was calculated using energy conservation principles as described in Equation 13. The moment of inertia (I) was first calculated using Equation 5. Accounting for 70% energy transfer efficiency from the falling mass to the glass, the expected drop height was determined and converted to inches for use with the drop tower apparatus. This calculation was incorporated into the Excel spreadsheet with all variables clearly labeled with appropriate units.

Strain Gauge Mounting

In this lab, strain gauges were mounted onto the glass substrate and wired for future connection to a Wheatstone bridge circuit. The goal of this experiment was to practice the proper surface preparation, strain gage alignment, adhesive bonding, and soldering techniques required for accurate strain measurements in subsequent tests.

A single foil strain gauge was bonded to the surface of a glass phone screen using M-Bond 200 adhesive and catalyst. The glass surface was first cleaned with acetone to remove any contaminants. The gauge was positioned using PCT installation tape to ensure proper orientation along the principal strain direction. Catalyst was applied to promote adhesion, followed by a thin layer of M-Bond 200 adhesive. Firm pressure was applied during curing to achieve a uniform bond. After curing, the mounting tape was removed, and the gage installation was inspected for uniform adhesion and absence of air bubbles.

After the strain gauge was mounted, electrical leads were soldered to the gage tabs to allow for circuit connection. Two 26 AWG stranded wires were tinned and attached to the gage solder pads using minimal heat to prevent damage to the gage or glass substrate. These wires were then spliced and soldered to two color-coded 22 AWG solid wires that would later interface with a breadboard and data acquisition system. Electrical tape was applied to relieve mechanical stress on the solder joints.

Table 1: List of equipment used in Strain Gage Mounting

Equipment	Model Number/Description
Acetone	Solvent used for surface cleaning before mounting
Gauze	Lint-free, used to apply and remove cleaning agents
Tweezers	Reverse/self-closing type, used to ensure precise gage placement
Strain gage	Foil strain gage
PCT Gage Installation Tape	Transparent adhesive tape for positioning the gage
Applicator brush	To apply adhesive

M-Bond 200 Catalyst	For adhesive curing
M-Bond 200 Adhesive	Quick-curing adhesive for gage bonding to the glass surface

Table 2: List of Equipment used in Soldering Wires to Strain Gages Mounted on Glass

Equipment	Model Number/Description
24-inch 26awg stranded wire	x2, to connect to the strain gage solder pads
3-inch 22awg solid wire	x2, differing colors, to connect to the breadboard
Wire strippers	To remove insulation from the ends
Solder	Electrical solder to connect wires
Pliers	Optional, to hold wires during soldering
Electrical tape	To insulate the soldered joint and secure wires

Circuitry Guidelines

In this lab, the project circuitry was designed. This was split into three components: the Wheatstone Bridge, the Differential Amplifier, and the Low Pass Filter. First, the Wheatstone bridge was constructed on the breadboard. Four resistors, with the same nominal resistance, were arranged in a diamond configuration. The resistors that are not in contact with the strain gage will have a resistance that is equal to 120 ohms, which is the resistance of the strain gages. Then, a 5V DC supply was applied to the bridge, and the digital multimeter was attached to the midpoints of the bridge to measure the output from the Wheatstone bridge.

In the second portion of the lab, the differential amplifier was built, using the operational amplifier and four more resistors. The two sets of resistor pairs were matched to reduce any imbalances in the circuit. Additionally, the resistors were all greater than 1kOm to prevent the drawing of excessive current. The amplifier was tested by applying a differential voltage from the function generator and measuring the output. If there was a significant DC offset, the resistors were switched out to see if there was improvement.

Next, the active low-pass filter was built into the breadboard using an additional op-amp, two resistors, and a capacitor. This filter was supposed to provide an additional gain while also minimizing any unnecessary frequencies. The filter's cutoff frequency was calculated and implemented to avoid aliasing in the data that was sampled using the DAQ.

After building and testing each individual component, the three portions were combined into a single system. The Wheatstone bridge is connected to the input of the differential amplifier which is then fed into the active low-pass filter. The output was connected to the DAQ system to record data. Next, a gentle load was placed on the beam to generate strains and to collect data.

Calibration and Testing

This lab is a continuation of the previous lab where the project circuitry was built. Therefore, it involved calibrating the final circuit to ensure that the readings that the team was receiving were correct. Specifically, the team needed to calibrate the data to ensure that the voltage output caused by forces applied to the phone screen were reading as expected. In order to do this, the strain gauges were connected to the DAQ hardware system. Then, matlab code from a previous lab was altered to record the sensor output properly. The expected voltage output was set to a maximum measurement of 10 volts. Additionally, the program was set to record data over a period of 15 seconds when it was calibrating.

The phone screen was mounted onto a block with supports on both ends. First, a weight of 200 grams was placed on the screen, followed by a 500 gram weight, and finally a combination of both to get a total of 700 grams. This data was plotted using the DAQ software and the Matlab code previously mentioned. The voltages in the three ranges can be averaged and plotted to determine the static sensitivity of the system, which is the slope of the line. This value can then be compared with the theoretical static sensitivity to complete the calibration of the final system.

Table 3: List of equipment used in Project Calibration and Data Acquisition

Equipment	Description
DAQ System	National Instruments (NI) USB-6008
Mass 1	200 g
Mass 2	500 g
Phone Screen with strain gages	From previous labs
Breadboard with project circuitry	From previous labs
Wooden block	Used to mount and hold the screen in place for force application

Measuring Stress during Ball Drop Test

On the day of testing, the strain gauges were calibrated using the matlab code and process from the previously described. Once this was done, the block with the phone screen was placed at the bottom of the apparatus. There was a pipe, filled with bolt holes at different heights, placed over the block. First, the bolt was secured at a height close to the screen. Then, the ball was inserted into the pipe. Figure 1 below shows this process with the team’s completed circuit while measuring.

Figure 1: Image depicting the Testing Process

Next, a team member began to run the Matlab code. As soon as the data acquisition started, the bolt was removed, forcing the ball to fall onto the phone screen. This same process was repeated at different heights, until the phone screen shattered. For each trial, the matlab code was run again to ensure that the team recorded all of the data.

Summary of Results

The figure below shows our calibration curve graph on the testing day.

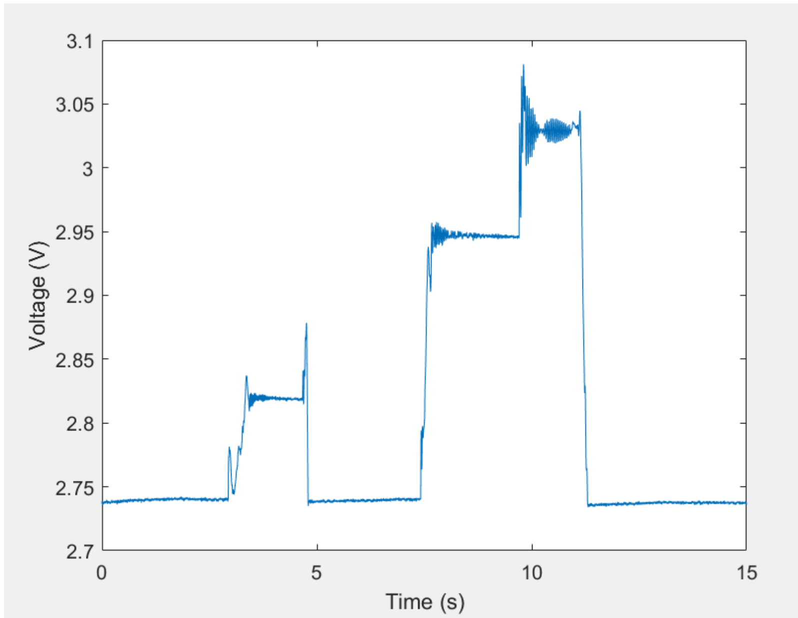

Figure 2: Sensor Calibration

As referenced in our procedure, after the conclusion of our calibration, we proceeded with the ball drop test to failure. Figure 3 shows the first ball drop at 25cm, in which the glass did not fail.

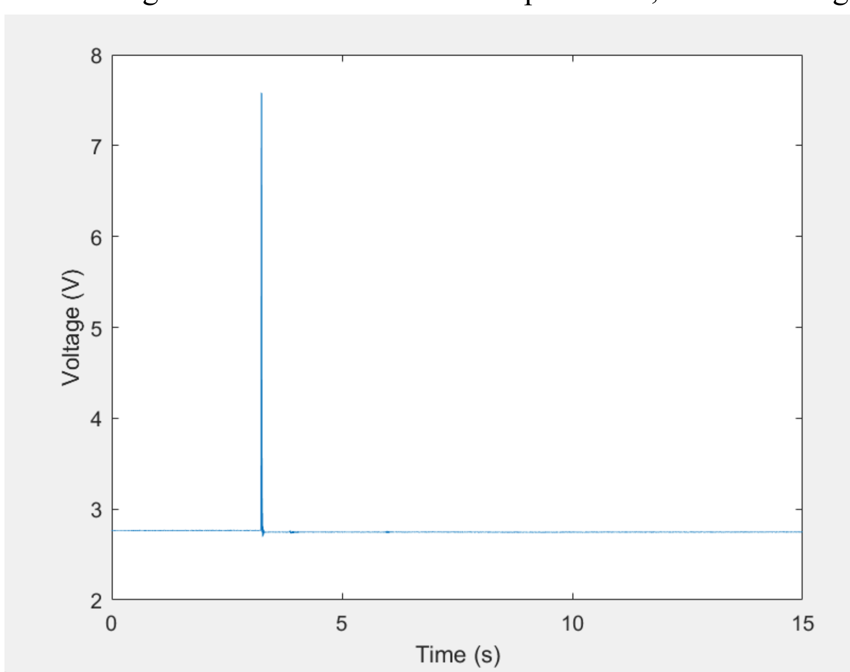

Figure 3: Sensor during 25cm ball drop

This is the second ball drop of 30 in, in which our glass experienced failure and shattered. Below Figure 4, we have Figure 5 that shows a further zoomed-in time of actual impact.

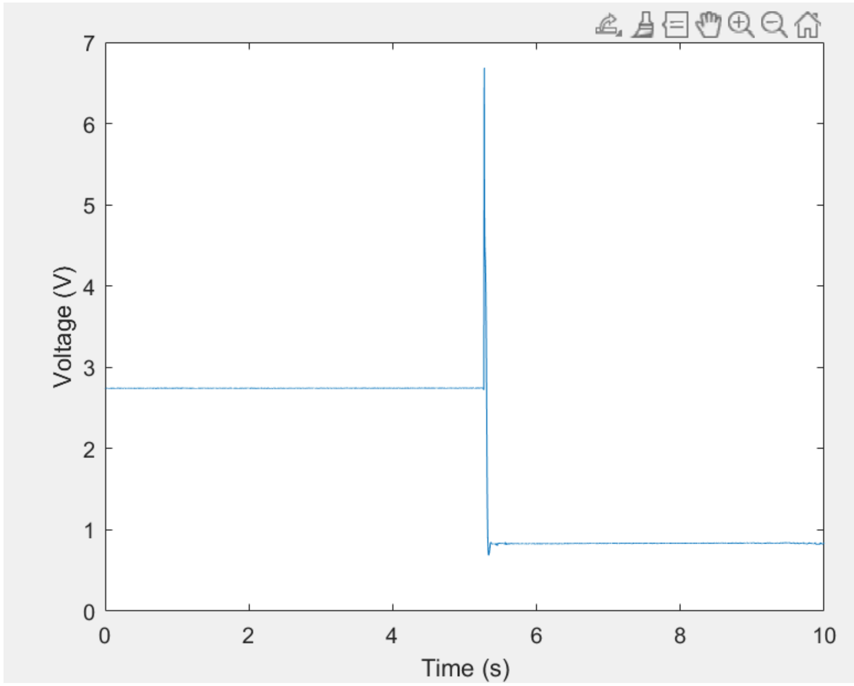

Figure 4: Sensor during 30cm ball drop

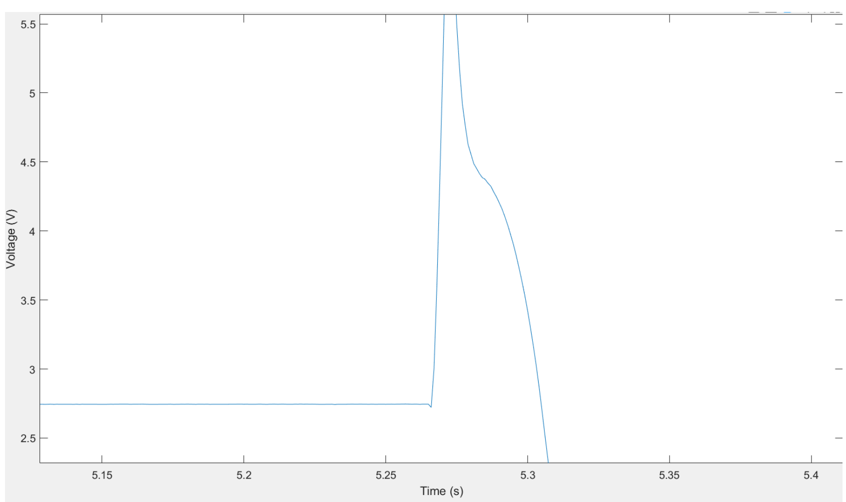

Figure 5: Zoomed in time of failure

The expected breakage stress from the ultimate stress of the glass was taken as 350 MPa ± 40 MPA as specified in the design requirements. To determine the static sensitivity from the calibration data, the average voltage for each applied mass was extracted from the plateau regions of the calibration curve (Figure 3). The masses were then converted to forces using $F = mg$, where $g = 9.81\ m/s^2$. A force vs. voltage table was created as shown in Table 4.

Table 4: Force vs Voltage Table

Force (N)	Voltage (V)
1.962	2.82
4.905	2.95
6.867	3.04

From the breakage test conducted at a drop height of 0.3m, the peak voltage at the moment of glass failure was measured as V = 6.68556 V. This voltage was converted to force using the calibration sensitivity and accounting for the 2.74 offset we had in calibration:

$$V = SF + b \Rightarrow F = \frac{V-b}{S} = \frac{6.68556 - 2.74}{0.0448} = 88.07\ N \qquad \text{[XVI]}$$

The breakage stress was then calculated using the beam bending equation for maximum stress at the center of a simply supported beam.

$$\sigma = \frac{M\frac{h}{2}}{I} = \frac{3*F*L}{2*b*h^2} = \frac{3*149.23*0.1}{2*0.064*0.00088^2} = 266.55\ \text{MPa} \qquad \text{[III]}$$

This measured breakage stress of 266.55 MPa is below the expected value of 350 MPa ± 40 MPa by approximately 24%.

The calculated expected breakage height was 0.345 m, based on the ultimate stress of 350 MPa and assuming 70% energy transfer efficiency from the falling mass to the glass beam. The actual breakage occurred at a drop height of 0.3 m. The actual breakage height was 13% lower than expected (0.3 m vs. 0.345 m). This appears to contradict the measured breakage stress of 266.55 MPa, which was 24% lower than the expected 350 MPa.

For the final step of the result section, t-test calculations were done with 7 samples from other members in the class. Table 5 includes all 7 samples and their respective breaking stress.

Table 5: Combined Test Results

Sample	Breakage Stress (MPa)
1	202.7
2	449.0
3	290.7
4	84.8
5	217.2

6	173.1
7	266.6

From the table, the sample mean is 240.59 MPa, with a standard deviation of 113.58 MPa and expected mean of 350 MPa. The significance level (α) = 0.01 with a 99% confidence, two-tailed. Then using these values within equation XIX, we calculated the t-statistic value.

$$t_{stat} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{(\frac{s_1}{n_1})^2 + (\frac{s_2}{n_2})^2}} = \frac{240.59 - 350}{\frac{113.58}{\sqrt{7}}} = -2.548 \qquad [XIX]$$

Therefore the |t-stat| = 2.548 < t-critical = 3.707. Conclusion: We fail to reject the null hypothesis at the 99% confidence level.

Discussion

1. Comparison to theory (calibration)
   From beam bending theory, the theoretical strain due to known forces can be calculated by equation [IV] and data from the Excel spreadsheet:
   $$\varepsilon = \frac{\sigma}{E} = \frac{3PL}{Ebh^2} = \frac{3(115.64)*0.1}{77 * 10^{\wedge}9(0.064)(0.00088^2)} = 9.091\ [\mu\varepsilon] \qquad [IV]$$
   In calibration, the expected strains are listed below, calculated from Table 4 and equation [IV]:

Table 6: Force vs Expected Strain Table

Force (N)	Expected Strain(με)
1.962	154
4.905	385
6.867	539

   However, the experimental behavior does not match these values exactly. The differences arise because the theoretical model assumes a perfectly simply supported beam, exact gauge placement, uniform material properties, and ideal strain transfer. In practice, support friction, small variations in glass thickness, gage misalignment, adhesive effects, and electrical offsets all cause the measured response to deviate from the expected strain. Consequently, the experimental system output is similar in trend but not identical in magnitude to the theoretical strain prediction.
2. Comparison of breakage stress
   The measured breakage stress for this sample is 266.55 MPa, which is about 24% lower than the nominal value of 350 MPa and roughly 14% below the lower bound of 310 MPa.

Thus, this particular glass sample is weaker than expected based on the design specification.

Based on the two-tailed t-test at a 99% confidence level, the results do not fully match the expectation, but do not differ significantly from the expected value. The deviation is reasonable for a brittle material like glass. Breakage stress is highly sensitive to surface flaws, edge chips, and handling damage, so different samples from the same batch can fail at very different stresses. Also, the bending stress model is quasi-static, while the actual test uses a dynamic impact, so there are effects like contact duration, local stress concentration, and potential vibration compared to the idealized theory.

The main sources of uncertainty include small surface defects or scratches on the glass, limited calibration points and voltage offsets, slight deviations in impact location, non-ideal or misaligned support conditions, DAQ sampling limitations, electrical noise or drift, and the strain gage placement being offset from the true beam center.

3. <u>Recommendations</u>

For the design, we would improve the strain gage mounting and configuration by placing the gages near the midspan where strain is highest, while laterally offsetting them to avoid contact with the impact point. Mounting the gauges on the back side of the glass could further prevent accidental collision. The bonding process would be more carefully controlled, like adhesive thickness, curing conditions, and surface preparation. In addition, more calibration points over a wider force range would help reduce uncertainty in the force-voltage relationship. Improved shielding and cable management would also be implemented to minimize electrical noise and drift.

For the entire test, we would make the impact conditions more controlled and repeatable. This could include a more precise drop-height adjustment with clear units, a guided impactor to ensure the ball always hits at the beam center, and more ideally, standardized supports to better approximate simple supports. A higher DAQ sampling rate or a short-time high-speed capture mode would help resolve the true peak voltage at break more accurately. We would also test multiple specimens under the same conditions to better separate sample-to-sample material variability from measurement error, and, if possible, pre-characterize the glass surface quality to avoid defects to better relate breakage stress to flaw conditions.

Conclusion

This experiment involved designing a sensor to measure the strength of a replacement iPhone screen to determine if the expected value listed by the Original Equipment Manufacturer was accurate. Using a ball drop test, the team was able to determine the ultimate breakage force. It was found that the measured breakage stress is 266.55 MPa, which is about 24% lower than the nominal value of 350 MPa and roughly 14% below the lower bound of 310 MPa.

Reference

[1] Whitefoot, J., “Project Lab 1: Detailed Gage Placement Design Strategy”, University of Pittsburgh, 2025.
[2] Whitefoot, J., “Project 2 lab - Strain Gage Mounting”, University of Pittsburgh, 2025.
[3] Whitefoot, J., “Project 3 - Circuitry Guidelines”, University of Pittsburgh, 2025.
[4] Whitefoot, J., “Project 4-5 - Calibration and Testing”, University of Pittsburgh, 2025.